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Commentary

Governing artificial intelligence in Indonesia digital health system: interoperability, equity, and policy readiness

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Artificial intelligence (AI) is entering health systems at a moment when governments are under pressure to expand access, improve quality, contain expenditure and make better use of health data. In Indonesia, this policy window is particularly important. The country is advancing digital health transformation through electronic medical records, national data integration and the SATUSEHAT (Indonesian for 'one health') platform [1], whilst AI is increasingly framed as a tool for clinical decision support, population health surveillance, administrative efficiency and personalised care. The central policy question, however, is not whether AI should be used in health care. It is whether Indonesia health system is sufficiently prepared to govern AI safely, equitably and accountably before its use becomes routine.

This commentary argues that Indonesia should treat AI in health not as a stand-alone technological innovation, but as a governance challenge within the wider digital health ecosystem. Interoperability, equity and health policy readiness should therefore be used as three practical tests of responsible health-AI adoption: whether AI systems can connect safely with standardised, auditable and clinically meaningful health data; whether their benefits and risks are fairly distributed across populations, regions and facilities; and whether institutions can validate, monitor, correct and hold AI systems accountable across their lifecycle. Without these tests, AI risks becoming another layer of digital complexity placed on top of uneven infrastructure, fragmented data quality, variable institutional capacity and unresolved public trust.

The timing is critical. Globally, policy discussions have moved from AI potential to AI implementation. WHO guidance has emphasised that health AI must be governed through principles of autonomy, safety, transparency, accountability, inclusiveness and sustainability [2]. More recent guidance on large multi-modal models extends this concern to generative AI applications that may support diagnosis, triage, documentation, medical education, patient communication and public health decision-making [3]. The OECD has similarly warned that the scaling of AI in health is constrained not only by technical capability, but also by fragmented data foundations, regulatory uncertainty, governance gaps and workforce capacity [4]. For Indonesia, these concerns are not abstract. They intersect directly with the country ongoing efforts to standardise, exchange and use individual health data at national scale.

A recent assessment of AI governance readiness in ASEAN found substantial variation across countries and highlighted the need for stronger country-level policies, digital infrastructure and human capital to support AI adoption in health care [5]. Indonesia appears in this regional picture as a country with policy momentum but unfinished implementation capacity. That creates both opportunity and risk. The opportunity is that Indonesia can embed AI governance early within its digital health architecture. The risk is that AI applications may spread through hospitals, primary care facilities, telemedicine providers, insurers, laboratories and private vendors before clear rules exist for validation, monitoring, accountability and redress.

The first policy issue is interoperability. In digital health, interoperability is often discussed as the technical capacity of systems to exchange data. For AI, this definition is too narrow. AI-ready interoperability must include data standardisation, data provenance, metadata quality, representativeness, auditability and clinical meaning. A technically connected system can still produce unsafe AI outputs if its data are incomplete, biased, poorly coded, outdated or generated under inconsistent clinical workflows. AI does not merely 'use' health data; it learns from the assumptions, omissions and inequalities embedded in those data.

This distinction matters for Indonesia because national integration platforms can create the impression that once data flow, AI readiness follows. That assumption should be resisted. Data integration is a necessary foundation, but it is not sufficient for AI governance. An AI model trained on fragmented or non-representative data may appear statistically powerful whilst performing poorly for populations, regions or facilities under-represented in the dataset. In a geographically diverse country, this can translate into uneven performance across urban and

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rural areas, public and private facilities, or populations with different levels of digital access.

The second issue is equity. AI can improve health equity if it supports earlier detection, better triage, more efficient referrals and stronger public health surveillance. Yet it can also widen inequity if deployed in settings where infrastructure, data quality, digital literacy and clinical supervision are uneven. Bias in health AI is not only a technical defect; it is a policy outcome. It can arise from who is included in datasets, which clinical problems are prioritised, whose language is represented, which facilities can adopt tools, and whether patients have meaningful ways to question AI-supported decisions.

Recent literature shows that bias can emerge across the entire AI lifecycle, from problem formulation and data collection to deployment and post-market surveillance [6]. This is highly relevant to Indonesia. If AI tools are developed mainly from data-rich tertiary hospitals, they may not generalise to under-resourced primary care settings. If tools are designed around urban digital users, they may underserve older adults, rural populations, people with disabilities, or patients with limited digital literacy. If AI procurement prioritises efficiency metrics alone, it may reward tools that reduce administrative burden without improving fairness, safety or patient outcomes.

The third issue is policy readiness. Health-AI readiness should not be equated with the existence of a national AI strategy or digital health platform. Readiness requires institutions capable of evaluating, authorising, monitoring and correcting AI systems throughout their lifecycle. The FUTURE-AI consensus guideline emphasises fairness, universality, traceability, usability, robustness and explainability as principles for trustworthy and deployable health AI [7]. These principles are useful, but they need translation into policy instruments that fit Indonesia health system.

Indonesia should therefore move towards a national health-AI governance framework built around five functions.

First, AI use cases should be risk-classified according to their clinical and public health consequences. Administrative AI, patient-facing chatbots, diagnostic support, triage tools, treatment recommendation systems and population surveillance models should not face the same governance requirements. The greater the potential effect on diagnosis, treatment, prioritisation or access, the stronger the requirements for evidence, human oversight and accountability.

Second, health-AI evaluation should require local validation. Imported or vendor-developed tools should not be assumed safe because they perform well elsewhere. Local validation should assess performance across demographic groups, facility types, regions, languages and clinical contexts. For high-risk tools, validation should include prospective evidence and post-deployment monitoring, not merely retrospective accuracy claims.

Third, Indonesia needs an explicit mechanism for algorithmic accountability. Every AI system used in health care should have identifiable institutional owners, documented intended use, version control, audit trails, escalation procedures and mechanisms for reporting harm or near misses. The problem of ‘shadow AI’, informal or unapproved AI use outside organisational oversight, is increasingly recognised internationally and should be anticipated before it becomes normalised in Indonesian clinical practice [8].

Fourth, governance must include patients and health workers. AI implementation will fail if it is perceived as imposed from technology vendors or central authorities without adequate clinical participation. Clinicians need training not only to operate AI tools, but also to understand their limitations, challenge outputs and communicate uncertainty to patients. Patients need clear information on when AI is used, what data are processed, what decisions remain human, and how they can seek explanation or correction.

Fifth, health policy should embed equity metrics into AI adoption. The success of health AI should not be measured only by speed, accuracy, coverage or cost reduction. It should also be assessed by whether it narrows or widens disparities in access, diagnosis, waiting time, referral quality and patient safety. Equity-sensitive monitoring would make Indonesia health-AI policy more aligned with universal health coverage and the ethical obligations of public health.

A practical way forward is to establish a national health-AI regulatory sandbox connected to SATUSEHAT and electronic medical record implementation. Such a sandbox should not be a deregulated playground for innovation. It should be a controlled policy environment where developers, health facilities, regulators, clinicians, patient representatives and data protection authorities jointly test AI tools under defined risk, evidence and accountability conditions. This would allow Indonesia to encourage innovation whilst avoiding premature, poorly monitored deployment.

The novelty of Indonesia health policy challenge lies in sequencing. Many countries built AI governance after AI applications had already entered clinical environments. Indonesia still has the opportunity to place governance upstream. By connecting AI oversight to digital health interoperability, the country can avoid treating AI safety, data protection, clinical validation and equity as separate policy conversations. They are the same conversation viewed from different points in the health system.

For Health Policy and Technology readers, Indonesia offers a relevant case because it illustrates the policy transition from digital health infrastructure to algorithmic health governance. The lesson is transferable: national data platforms may enable AI, but they do not by themselves make AI trustworthy. Trustworthy health AI requires institutions that can govern data quality, clinical use, equity impact, vendor accountability and public legitimacy.

Indonesia digital health transformation should therefore be judged not by how rapidly AI tools are adopted, but by whether they are governed before they scale. In health care, premature scale can convert small design weaknesses into systemic risks. A responsible Indonesian pathway would put interoperability, equity and policy readiness at the centre of AI governance. This is not a brake on innovation. It is the condition under which innovation can safely become part of everyday care.

Use of artificial intelligence

AI-assisted language tools (Paperpal & Grammarly) were used solely for grammar and stylistic editing under the author’s direction. The author verified all content and accepts full responsibility for the accuracy and integrity of the manuscript. No AI system was used to generate scientific claims, analyse data, or draw conclusions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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